

Report No:
EE25-0437

ISSUE
B

DATE
4th February 2026

ARCHITECT
Archipelago

CLIENT
Brisbane Airport Corporation

Skygate Play

**Brisbane Airport Corporation The
Cct, Brisbane Airport, QLD**



**SCI QUAL
INTERNATIONAL**



**SCI QUAL
INTERNATIONAL**



**SCI QUAL
INTERNATIONAL**



Member of:



NCC Section J Report
Deemed to Satisfy Method

Energy Efficiency Services

**Ashburner Francis
Consulting Engineers**

Ashburner Francis Consulting Engineers Pty Ltd
977 Stanley St East, East Brisbane, QLD 4169
Phone: 07 3510 8888
Fax: 07 3510 8899
Email: brisbane@ashburnerfrancis.com.au
Web: www.ashburnerfrancis.com.au

Offices

Australia
Brisbane, Townsville, Toowoomba, Perth, Darwin

Copyright

All rights reserved. No part of the content of this report may be published, reproduced, transmitted or modified in any form or by any means without written permission from Ashburner Francis Pty Ltd.

Disclaimer

The building(s) was assessed using drawings listed in the Appendix that were current during the period of analysis. Any minor or major changes to the building envelope after the issue of this report may forfeit the results, due to input variables requiring amendments in calculations

QA System

Project Engineer: Lara Harris
Verification: Satya Vankadari

Revision History

Issue	Date	Description	Initials
A	11 th September 2025	Revision A	LH
B	4 th February 2026	Revision B	LH

TABLE OF CONTENTS

Introduction 4

Part J0 Energy Efficiency 6

Part J4 Building Fabric 7

Part J5 Building Sealing 14

Part J6 Air Conditioning and Ventilation Systems 16

Part J7 Artificial Lighting and Power 17

Part J8 Heated water supply and swimming pool and spa pool plant 18

Part J9 Energy monitoring and on-site distributed energy resources 19

Appendix – Referenced Drawings 20

INTRODUCTION

Brisbane Airport Corporation is proposing to develop a play centre building on their site located in Brisbane Airport.

The objective of the National Construction Code (NCC) 2022 section J is to reduce greenhouse gas emissions by efficiently using energy, by considering the building materials, glazing, lighting and ventilation.

A building solution that is proposed to comply with the Deemed-to-Satisfy (DTS) provisions satisfies the Performance Requirement JP1 by complying with J0 – J8.

This report details the DTS provisions applicable to the development and identifies measures to achieve compliance with the Deemed-to-Satisfy provisions. Measures identified to achieve compliance with DTS provisions are highlighted in blue.

Project Details

Building Details	Skygate Play
	The Cct, Brisbane Airport
NCC Building Classification	Class 9b
NCC 2022 Climate Zone	2

Refer to the drawings indicated in the appendix A for further details.

Location

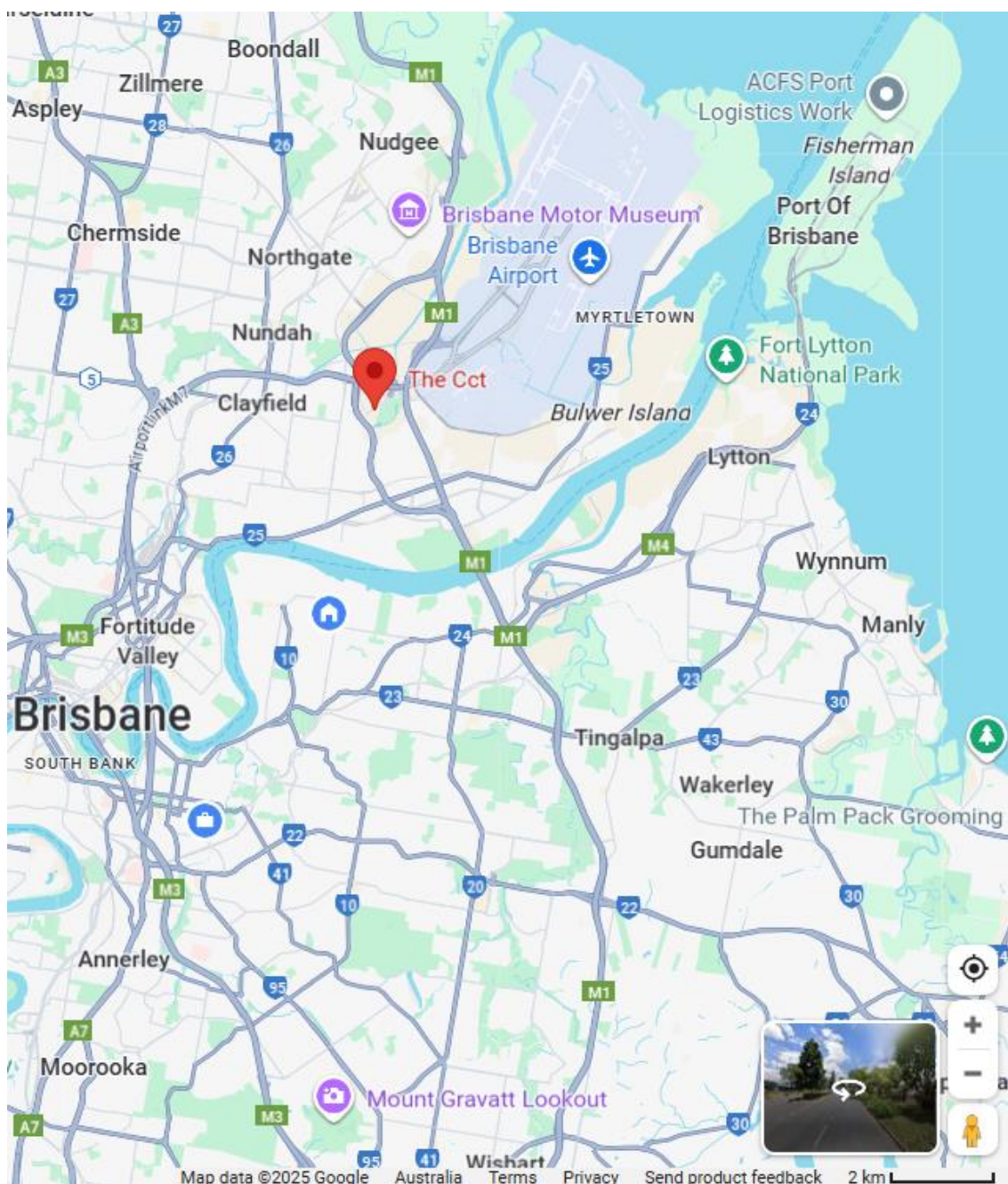


Figure 1 Locality Plan

PART J0 ENERGY EFFICIENCY

J0.1 Application of Section J

According to the NCC 2022:

- (1) For a Class 2 to 9 building, other than a sole-occupancy unit of a Class 2 building or a Class 4 part of a building, Performance Requirement J1P1 is satisfied by complying with—
 - (a) Part J4, for the building fabric; and
 - (b) Part J5, for building sealing; and
 - (c) Part J6, for air-conditioning and ventilation; and
 - (d) Part J7, for artificial lighting and power; and
 - (e) Part J8, for heated water supply and swimming pool and spa pool plant; and
 - (f) J9D3, for facilities for energy monitoring.
- (2) For a sole-occupancy unit of a Class 2 building or a Class 4 part of a building, Performance Requirement J1P2 is satisfied by complying with—
 - (a) J3D3, using house energy rating software; or
 - (b) the following—
 - (i) J3D4, for ceiling fans; and
 - (ii) J3D5, J3D6, J4D3, J4D7(3), J4D7(4) and Part J5, for general thermal construction; and
 - (iii) J3D7, for roofs; and
 - (iv) J3D8 and J3D11 to J3D13, or J3D9, for walls and glazing; and
 - (v) J3D10, for floors.
- (3) For a sole-occupancy unit of a Class 2 building or a Class 4 part of a building, Performance Requirement J1P3 is satisfied by complying with—
 - (a) for the net equivalent energy usage—
 - (i) J3D14, for a sole-occupancy unit of a Class 2 building or a Class 4 part of a building with a total floor area not greater than 500 m²; or
 - (ii) J3D15, using house energy rating software; and
 - (b) Part J6, for air-conditioning and ventilation; and
 - (c) Part J7, for artificial lighting and power.
- (4) For a Class 2 to 9 building, Performance Requirement J1P4 is satisfied by complying with J9D4 and J9D5.

Therefore, Parts J4 – J8 and J9D3-5 are applicable to the proposed building.

PART J4 BUILDING FABRIC

J4D2 Application of Part

According to the NCC 2022:

The Deemed-to-Satisfy Provisions of this Part apply to building elements forming the envelope of a Class 2 to 9 building other than J4D3(5), J4D4, J4D5, J4D6 and J4D7 which do not apply to a Class 2 sole-occupancy unit or a Class 4 part of a building.

Therefore, Part J4 applies to the proposed building.

J4D3 Thermal Construction General

According to the NCC 2022 J4D3:

- (1) Where required, insulation must comply with AS/NZS 4859.1 and be installed so that it—
 - (a) abuts or overlaps adjoining insulation other than at supporting members such as studs, noggings, joists, furring channels and the like where the insulation must be against the member; and
 - (b) forms a continuous barrier with ceilings, walls, bulkheads, floors or the like that inherently contribute to the thermal barrier; and
 - (c) does not affect the safe or effective operation of a service or fitting.
- (2) Where required, reflective insulation must be installed with—
 - (a) the necessary airspace to achieve the required R-Value between a reflective side of the reflective insulation and a building lining or cladding; and
 - (b) the reflective insulation closely fitted against any penetration, door or window opening; and
 - (c) the reflective insulation adequately supported by framing members; and
 - (d) each adjoining sheet of roll membrane being—
 - (i) overlapped not less than 50 mm; or
 - (ii) taped together.
- (3) Where required, bulk insulation must be installed so that—
 - (a) it maintains its position and thickness, other than where it is compressed between cladding and supporting members, water pipes, electrical cabling or the like; and
- (4) in a ceiling, where there is no bulk insulation or reflective insulation in the wall beneath, it overlaps the wall by not less than 50 mm.
Roof, ceiling, wall and floor materials, and associated surfaces are deemed to have the thermal properties listed in Specification 36.
- (5) The required Total R-Value and Total System U-Value, including allowance for thermal bridging, must be—
 - (a) calculated in accordance with AS/NZS 4859.2 for a roof or floor; or
 - (b) determined in accordance with Specification 37 for wall-glazing construction; or
 - (c) determined in accordance with Specification 39 or Section 3.5 of CIBSE Guide A for soil or sub-floor spaces.

Provided that the building fabric of the proposed buildings is constructed in accordance with NCC 2022 Part J4D3 (1), (2), (3) & (4) the building will comply with NCC 2022 Part J4D3.

J4D4 Roof and Ceiling Construction

In accordance with the NCC 2022, the minimum R-value for the roof and ceiling is as follows:

- (a) in climate zones 1, 2, 3, 4 and 5, R3.7 for a downward direction of heat flow; and
- (b) in climate zone 6, R3.2 for a downward direction of heat flow; and
- (c) in climate zone 7, R3.7 for an upward direction of heat flow; and
- (d) in climate zone 8, R4.8 for an upward direction of heat flow.

The proposed building is located in climate zone 2, and therefore a minimum R-value for roof and ceiling of R3.7

The solar absorptance of the upper surface of a roof must be not more than 0.45.

Roof Type

The roof is to be constructed of metal sandwich panel roof.

Roof: Metal roof ceiling	R Value [m²K/W]
Solar Absorptance ≤ 0.45	
Not more than 3 m/s wind	0.03
Kingspan KS1000RW 100mm	4.81
On a surface with a pitch not more than 5°	0.15
Total Value	4.99
Required Value	3.70
	PASS

The proposed external roof and ceiling structures comply with the current NCC 2022 DTS provisions provided that they are installed as detailed on the Architectural drawings referred to in the appendix and include the above mentioned insulation in the roof.

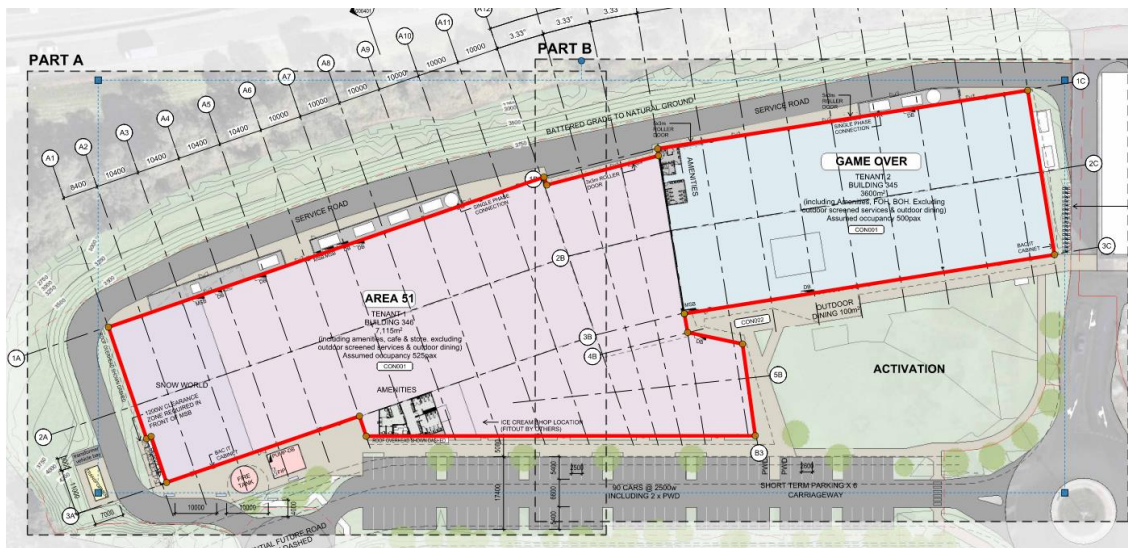
J4D5 Roof Lights

There are no roof lights in the proposed building, and therefore this clause does not apply.

J4D6 Walls & Glazing

Section J4D6 of the NCC 2022, is applicable to walls that form part of the envelope. This does not necessarily limit the requirements to external walls, but rather walls that are to a non-conditioned space subject to the environment outside.

The plans below indicate walls considered to be part of the buildings' envelope that must comply with Part J4D6.

**Display Glazing**

Display glazing is not indicated on the proposed building, and therefore this clause does not apply.

Walls construction

Wall component of a wall-glazing construction must achieve a minimum Total R-Value of –

- (a) Where the wall is less than 80% of the area of the wall-glazing construction R1.0; or
- (b) Where the wall is 80% or more of the area of the wall-glazing construction, the value specified in Table J4D6a.

Climate Zone	Class 2 Common area, Class 5, 6, 7, 8 or 9b building or class 9a building other than a ward area	Class 3 or 9c building or Class 9a ward area
1	2.4	3.3
2	1.4	1.4
3	1.4	3.3
4	1.4	2.8
5	1.4	1.4
6	1.4	2.8
7	1.4	2.8
8	1.4	3.8

The solar admittance of externally facing wall-glazing construction must not be greater than –

- (a) For a Class 2 common area, a Class 5, 6, 7, 8 or 9b building or Class 9a building other than a ward area the values specified in Table J4D6b; and
- (b) For a class 3 or 9c building or a class 9a ward area, the values specified in Table J4D6c

As the proposed Skygate Play is located in climate zone 2, the South & West walls have more than 80% of the area of the wall-glazing construction, and therefore a minimum R-value of 1.4. the North & East wall has a wall to glazing ratio of less than 80% as such the minimum R-value is 1.0

Wall Types

External wall: Tilt Panel clad 90mm stud
Not more than 3 m/s wind
Tilt Panel
Non-reflective air space
R2.0 Bradford Gold wall batts (90 mm)
Gypsum plasterboard
Internal wall

External wall: FC clad 90mm stud
Not more than 3 m/s wind
FC cladding
Non-reflective air space
R2.0 Bradford Gold wall batts (90 mm)
Gypsum plasterboard
Internal wall

External wall: Metal Sandwich panel
Not more than 3 m/s wind
Kingspan KS1000RW 100mm
Internal wall

Please note the above insulation thickness values and corresponding R-values are based on Bradford Gold® Wall Batts and Kingspan sandwich panel. Other types of insulation may be used to suit the application, however, to comply with this document, the insulation's R-value must be equal to or greater than that mentioned.

The proposed wall structures comply with the current NCC 2022 DTS provisions provided that they are installed as detailed on the Architectural drawings referred to in the Appendix and include the above mentioned insulation.

Glazing

In accordance with the NCC 2022, the total system U-Value for the wall-glazing construction must not be greater than:

- (a) For a Class 2 common area, a Class 5, 6, 7, 8 or 9b building or a Class 9a building other than a ward area, U2.0; and
- (b) For a Class 3 or 9c building or a Class 9a ward area –
 - (i) In climate zone 1, 3, 4, 6 or 7, U1.1; or
 - (ii) In climate zone 2 or 5, U2.0; or
 - (iii) In climate zone 8, U0.9.

As the proposed Skygate Play is class 9b building located in climate zone 2, and therefore maximum total system U-value for the wall-glazing construction of U2.0.

The glazing for the Skygate Play at must not exceed the allowance for the aggregate air conditioning energy value given for the particular façade.

The analysis conducted used U-values (NFRC) for the glazing and frame obtained from the WERS (Windows energy rating scheme) website for:

Framing	Glazing	U-Value	SHGC
G James 450 Aluminium Fixed Window	6mm Eclipse Advantage Clear	4.8	0.54
G James Type 472 Aluminium Hinged door	6mm Eclipse Advantage Clear	4.7	0.47
G James Type 477-570 Aluminium Bifold door	6mm Eclipse Advantage Clear	4.6	0.49
Sliding door	6mm Eclipse Advantage Clear	4.6	0.49
Danpalon	8mm Grey / Opal	3.3	0.45

The U-Value and SHGC indicated above are the whole window values for the framing and glazing. Other types of glazing systems may be used to suit the application, however, to comply with this document, the total glazing systems U-value and SHGC value must be equal to or better than that mentioned. Please note higher efficiency glazing has U-Values and SHGC values closer to zero.

The total system U-value and solar admittance of wall-glazing construction must be calculated in accordance with NCC2022 specification J4D6a and this is calculated through the façade calculator.

Calculator

Façade Report

Calculator

Project Summary

Date
4/02/2026

Name
Lara Harris

Company
Ashburner Francis

Position
Director

Building Name / Address
Skygate Play
Area 51 & Game Over

Building State
QLD

Climate Zone
Climate Zone 2 - Warm humid
summer, mild winter

Building Classification
Class 9b - public halls,
function rooms or the like

Stores Above Ground
1

Tool Version
1.2 (June 2020)

The summary below provides an overview of where compliance has been achieved for Specification J1.5a - Calculation of U-Value and solar admittance - Method 1 (Single Aspect) and Method 2 (Multiple Aspects).

Compliant Solution =
Non-Compliant Solution =

	North	East	Method 1 South	West	Method 2 All
Wall-glazing U-Value (W/m².K)	1.50	1.36	0.42	0.42	0.96
Solar Admittance	0.09	0.12		0.01	
AC Energy Value					508

Method 1

Method 1 charts show Wall-glazing U-Value (W/m².K) and Solar Admittance (SA) for North, East, South, and West. The charts compare the Proposed Design (solid bars) against the DTS Reference (dashed lines). For North, U-Value is 1.50 and SA is 0.085. For East, U-Value is 1.36 and SA is 0.122. For South, U-Value is 0.42 and SA is 0.01. For West, U-Value is 0.42 and SA is 0.01.

Method 2

Method 2 charts show Wall-glazing U-Value (W/m².K) and AC Energy (J/m²) for the Proposed Design (solid bars) against the DTS Reference (dashed lines). For U-Value, the Proposed Design is 0.96 and the DTS Reference is 2.00. For AC Energy, the Proposed Design is 508 and the DTS Reference is 650.

Project Details				
	North	East	South	West
Glazing Area (m²)	462.0011	346.32035	0	15
Glazing to Façade Ratio	32%	25%	0%	2%
Glazing References	FW-1 Dampson -1 SD-1 Dampson - colour 1 HD-1 FW-2	FW-1 HD-1 Dampson -1 FW-2 HD-2 BFD-1 SD-1 Dampson-2		Dampson -1
Glazing System Types	0	0		0
Glass Types	Single glazing	Single glazing		Single glazing
Frame Types	Aluminium	Aluminium		
Average Glazing U-Value (W/m².K)	3.55	4.20		3.30
Average Glazing SHGC	0.41	0.49	0.00	0.45
Shading Systems	Horizontal	Horizontal	Horizontal	Horizontal
Wall Area (m²)	961.83	1060.58	1472.14	942.49
Wall Types	Wall	Wall	Wall	Wall
Methodology	Wall			
Wall Construction	FC ag ntb R2 pb TP ag ntb R2 pb KS1000 100mm	FC ag ntb R2 pb TP ag ntb R2 pb KS1000 100mm	KS1000 100mm TP ag ntb R2 pb FC ag ntb R2 pb	KS1000 100mm TP ag ntb R2 pb FC ag ntb R2 pb
Wall Thickness	120 260 100	120 260 100	100 260 120	260
Average Wall R-value (m².K/W)	1.92	2.29	2.40	2.67
Solar Absorbance	0.6	0.6	0.6	0.6

IMPORTANT NOTICE AND DISCLAIMER IN RESPECT OF THIS CALCULATOR

By accessing or using this calculator, you agree to the following: Where used, this calculator, it may not be complete or up-to-date. You can ensure that you are using a complete and up-to-date version by checking the Australian Building Codes Board website (www.abcc.gov.au). The Australian Building Codes Board, the Commonwealth of Australia and States and Territories of Australia do not accept any liability, including liability for negligence, for any loss (however caused), damage, injury, expense or cost incurred by any person as a result of accessing, using or relying upon this calculator, to the maximum extent permitted by law. No representation or warranty is made or given as to the currency, accuracy, reliability, merchantability, fitness for any purpose or completeness of this calculator or any information which may appear on any third party materials, and all such representations and warranties are excluded to the extent permitted by law. This calculator is not legal or professional advice. Persons rely upon this calculator entirely at their own risk and must take responsibility for assessing the relevance and accuracy of the information in relation to their particular circumstances.

© Commonwealth of Australia and the States and Territories of Australia 2019, published by the Australian Building Codes Board.
The material in this calculator is licensed under a Creative Commons Attribution-Non-Commercial-4.0 International license, with the exception of third party materials and any trade marks. It is provided for general information only and without warranties of any kind. You may not make derivatives of this calculator, but may only use a verbatim copy. More information on this CC BY-NC license is set out at the Creative Commons website. (<http://www.creativecommons.org/licenses/by-nc/4.0/>). For information regarding this calculator, see www.abcc.gov.au.

J4D7 Floors

In accordance with the NCC 2022 Table J4D7, the total R-value for the floor of any building without in-slab heating or cooling system is:

- R2.0 for climate zone 1, 2, 3, 4, 5, 6 and 7.
- R3.5 for climate zone 8.

The NCC 2022 provides a clarification on the above point by stating.

- (2) For the purposes of (achieving the above mentioned R-values), a slab-on-ground that does not have an in-slab heating or cooling system is considered to achieve a Total R-Value of R2.0, except—
- (a) in climate zone 8; or
 - (b) a Class 3, Class 9a ward area or Class 9b building in climate zone 7 that has a floor area to floor perimeter ratio of less than or equal to 2.

As the proposed Skygate Play has a slab on ground without in-slab heating or cooling in climate zone 2, it is assumed to achieve a total R-value of R2.0.

Floor: On-ground	Thickness [mm]	R Value [m²K/W]
Internal floor airfilm		0.11
Solid concrete	200	0.14
Soil in contact with slab		2.00
Total Value		2.25
Required Value		2.00
		PASS

PART J5 BUILDING SEALING

J5D2 Application of Part

According to the NCC 2022:

The Deemed-to-Satisfy Provisions of this Part apply to elements forming the envelope of a Class 2 to 9 building, other than—

- (a) a building in climate zones 1, 2, 3 and 5 where the only means of air-conditioning is by using an evaporative cooler; or
- (b) a permanent building opening, in a space where a gas appliance is located, that is necessary for the safe operation of a gas appliance; or
- (c) a building or space where the mechanical ventilation required by Part F6 provides sufficient pressurisation to prevent infiltration.

Therefore, Part J5 applies to the proposed building.

J5D3 Chimneys and Flues

There are no chimneys or flues therefore, this clause does not apply.

J5D4 Roof Lights

- (1) A roof light must be sealed, or be capable of being sealed, when serving—
 - (a) a conditioned space; or
 - (b) a habitable room in climate zones 4,5,6,7 or 8.
- (2) A roof light required by (1) to be sealed, or capable of being sealed, must be constructed with —
 - (a) An imperforated ceiling diffuser or the like installed at the ceiling or internal lining level; or
 - (b) A weatherproof seal; or
 - (c) A shutter system readily operated either manually mechanically or electronically by the occupant.

As the roof light isn't open able it is assumed to provide a weatherproof seal therefore it complies with this clause.

J5D5 Windows and Doors

All external doors and windows, except those complying with AS2047, fire or smoke doors, roller shutter doors, security grilles or the like installed specifically for afterhours security, or in climate zones 4, 5, 6, 7 or 8 are required to be sealed.

- (3) A seal to restrict air infiltration—
 - (a) for the bottom edge of a door, must be a draft protection device; and
 - (b) for the other edges of a door or the edges of an openable window or other such opening, may be a foam or rubber compression strip, fibrous seal or the like.
- (4) An entrance to a building, if leading to a conditioned space must have an airlock, self-closing door, rapid roller door, revolving door or the like, other than—
 - (a) where the conditioned space has a floor area of not more than 50 m²; or
 - (b) where a café, restaurant, open front shop or the like has—
 - (i) a 3 m deep un-conditioned zone between the main entrance, including an open front, and the conditioned space; and
 - (ii) at all other entrances to the café, restaurant, open front shop or the like, self-closing doors.

- (5) A loading dock entrance, if leading to a conditioned space, must be fitted with a rapid roller door or the like.

J5D6 Exhaust Fans

All exhaust fans are required to be fitted with a sealing device such as a self-closing damper or the like when serving a conditioned space or a habitable room in climate zone 4, 5, 6, 7 or 8.

J5D7 Construction of Ceilings, walls and floors

Ceilings, walls, floors and any opening such as a window frame, door frame, roof light frame or the like must be constructed to minimise air leakage when forming part of the envelope or in climate zone 4, 5, 6, 7 or 8.

Construction must be enclosed by internal lining systems that are close fitting at ceiling, wall and floor junctions or sealed at junctions and penetrations with close fitting architrave, skirting, cornice, expanding foam, rubber compressible strip, caulking or the like.

J5D8 Evaporative coolers

There are no evaporative coolers therefore, this clause does not apply.

Provided that the building sealing is in accordance with the NCC 2022 parts:
J5D5 for sealing requirements for external windows and doors,
J5D6 for sealing requirements for exhaust fans,
J5D7 for sealing requirements for construction of roofs, walls and floors,
the buildings will comply with requirements for Building Sealing in accordance with NCC 2022 Part J5.

PART J6 AIR CONDITIONING AND VENTILATION SYSTEMS

J6D2 Application of Part

This part applies to all air-conditioning and ventilation systems in the building.

The contract documentation for the mechanical services installation should be installed in accordance with the energy efficiency requirements for air conditioning and ventilation systems set out in Part J6. These requirements applicable to the building include at minimum (this list is not exhaustive):

- Air conditioning systems must be capable of being inactivated when the part of the building is not occupied.
- Air conditioning units must comply with the Minimum Energy Performance Standard (MEPS) and the Queensland Development Code MP4.1.
- Air conditioning systems over 2kW_r of cooling or 1kW_r of heating and mechanical ventilation over 1000L/s must be time switched in accordance with Specification J6:
 - A time switch must be capable of switching on and off electric power to systems at variable pre-programmed times and on variable pre-programmed days.
- Insulation of ductwork and piping in accordance with NCC 2022 Specification J6D6 and J6D9 respectively.
- Mechanical ventilation must be capable of being inactivated when the part of the building is not occupied.

PART J7 ARTIFICIAL LIGHTING AND POWER

J7D2 Application of Part

This part applies to the artificial lighting and power in the building.

The electrical services installation must be designed in accordance with the energy efficiency requirements for artificial lighting and power set out in Part J7.

J7D3 Interior Artificial Lighting

The energy efficiency requirements for the artificial lighting within Skygate Play must be designed in compliance with section J7D3.

J7D4 Interior Artificial Lighting and Power Control

J7D4 (1) stipulates that all artificial lighting of a room must be individually switched or operated by another control device. The design shall comply with this clause.

J7D4 (2) only applies to Class 3 Building, and is therefore not applicable to these buildings.

J7D4 (3) stipulates the requirements for the switching of lights. J7D4 (3) (a). stipulates that light switches are to be located in a visible position, either in the room being switched or in an adjacent space where the lights are visible. J7D4 (3) (b). stipulates that for spaces less than 2000m² a light switch should not operate lighting within an area of more than 250m². For areas greater than 2000m², a light switch should not operate lighting within an area of more than 1000m². The design shall comply with this clause.

J7D4 (4) stipulates for building or storey of building that is more than 250m², 95% of the light fitting in the building must be controlled by a time switch or an occupant sensing device. The design shall comply with this clause.

J7D4 (5) stipulates the requirement of having separately switched artificial lighting in the natural lighting zone for a building Class 5, 6 or 8 of more than 250 m², therefore the design shall comply with this clause.

J7D4 (6) stipulates that motion detector in accordance with Specification J6 is required to control artificial lighting in fire-isolated stairways, fire-isolated passageway or fire-isolated ramp.

J7D4 (7) stipulates that artificial lighting in foyers, corridor and other circulation spaces require more than 250W within a single zone and to be adjacent windows.

J7D4 (9) emergency lighting compliant with Part E4 need not comply with Part J7D4 (1)-(8).

J7D4 (9) (b) only applies to building with 24 hour occupancy and is therefore not applicable.

J7D4 (10) stipulates that J7D4 (4) is not applicable to patient care areas in a Class 9a building or in a Class 9c aged care building, plant room, motor room, workshops where power tools are used and also a heater where the heater also emits light such as ion bathrooms and is therefore not applicable..

J7D5 Interior Decorative and Display Lighting

Interior decorative or display lighting must be switched separately from other lighting. If the lighting exceeds 1 kW, it must be controlled by a time switch. Window display lighting is to be controlled separately from other display lighting. There are no interior decorative and display lighting, therefore, clause J7D5 is not applicable.

J7D6 Exterior artificial lighting

The external lighting is required to be controlled by a daylight sensor or a time switch. If the total external lighting load exceeds 100W, it is required to be controlled by a motion sensor or use LED luminaries for 90% of the total lighting load. If used for decorative purposes, it must be on a separate time switch in accordance with Specification J7D6.

All external lights are controlled by a time switch/photo cell and selected to comply with the minimum requirement of W/Lumen and therefore complied with clause J7D6.

J7D7 Boiling Water and Chilled Water Storage Units

Power supplies to any chilled or boiling water storage unit must be controlled by a time switch.

J7D8 Lifts

Lifts must be configured to ensure artificial lighting and ventilation in the car are turned off when it is unused for 15 minutes. They need to achieve the idle and standby energy performance level and energy efficiency class in ISO 25745-2. There is no lift, therefore, clause J7D8 is not applicable.

J7D9 Escalators and moving Walkways

Escalators and moving walkways must have the ability to slow to between 0.2 m/s and 0.05 m/s when unused for more than 15 minutes. There are no escalators or moving walkways, therefore, clause J7D9 is not applicable.

PART J8 HEATED WATER SUPPLY AND SWIMMING POOL AND SPA POOL PLANT

J8D2 Heated water supply

A heated water supply system for food preparation and sanitary purposes must be designed and installed in accordance with Part B2 of NCC Volume Three — Plumbing Code of Australia.

J8D3 Swimming pool heating and pumping

There is no swimming pool within this project, therefore, clause J8D3 is not applicable.

J8D4 Spa pool heating and pumping

There is no spa pool within this project, therefore, clause J8D4 is not applicable.

PART J9 ENERGY MONITORING AND ON-SITE DISTRIBUTED ENERGY RESOURCES

J9D2 Application of Part

Part J9 is not applicable to Class 2 or Class 4 part buildings. Therefore, the Deemed-To-Satisfy provisions of Part J9 apply to Skygate Play.

J9D3 Facilities for energy monitoring

- (1) A building or sole-occupancy unit with a floor area of more than 500 m² must have energy meters configured to record the time-of-use consumption of gas and electricity.
- (2) A building with a floor area of more than 2 500 m² must have energy meters configured to enable individual time-of-use energy data recording, in accordance with (3), of—
 - (a) air-conditioning plant including, where appropriate, heating plant, cooling plant and air handling fans; and
 - (b) artificial lighting; and
 - (c) appliance power; and
 - (d) central hot water supply; and
 - (e) internal transport devices including lifts, escalators and moving walkways where there is more than one serving the building; and
 - (f) on-site renewable energy equipment; and
 - (g) on-site electric vehicle charging equipment; and
 - (h) on-site battery systems; and
 - (i) other ancillary plant.
- (3) Energy meters required by (2) must be interlinked by a communication system that collates the time-of-use energy data to a single interface monitoring system where it can be stored, analysed and reviewed.
- (4) The provisions of (2) do not apply to energy meters serving—
 - (a) a Class 2 building where the total floor area of the common areas is less than 500 m²; or
 - (b) individual sole-occupancy units with a floor area of less than 2 500 m².

APPENDIX – REFERENCED DRAWINGS

Architectural Drawings

SHEET NUMBER	SHEET NAME	REVISION
000000	COVER SHEET	8
000020	PERSPECTIVES	10
000101	SITE PLAN	10
000200	TENANCY PLAN	12
000201	GENERAL ARRANGEMENT - PART A	8
000202	GENERAL ARRANGEMENT - PART B	8
000221	STRUCTURAL SETOUT PLAN - PART A	2
000222	STRUCTURAL SETOUT PLAN - PART B	2
000241	REFLECTED CEILING PLAN - PART A	4
000242	REFLECTED CEILING PLAN - PART B	4
000251	ROOF PLAN - PART A	8
000252	ROOF PLAN - PART B	9
000301	ELEVATIONS 01	9
000302	ELEVATIONS 02	9
000303	ELEVATIONS 03	9
000401	SECTIONS 01	8
000402	SECTIONS 02	2
000800	WET AREA DETAILS	2
000802	WET AREA DETAIL - INTERNAL ELEVATIONS	1
000803	WET AREA DETAIL - INTERNAL ELEVATIONS	1